BE PRO

MODULE 2
CLASS 5

Calculus, Probability, and Statistics

MODULE 2.
PROGRAMMING AND MATHEMATICAL FOUNDATIONS

OVERVIEW

TOPICS TO COVER

~~1. Python Crash Course + Environment Setup~~ ✓

~~2. NumPy and Pandas~~ ✓

~~3. Data Visualization — Matplotlib and Seaborn~~ ✓

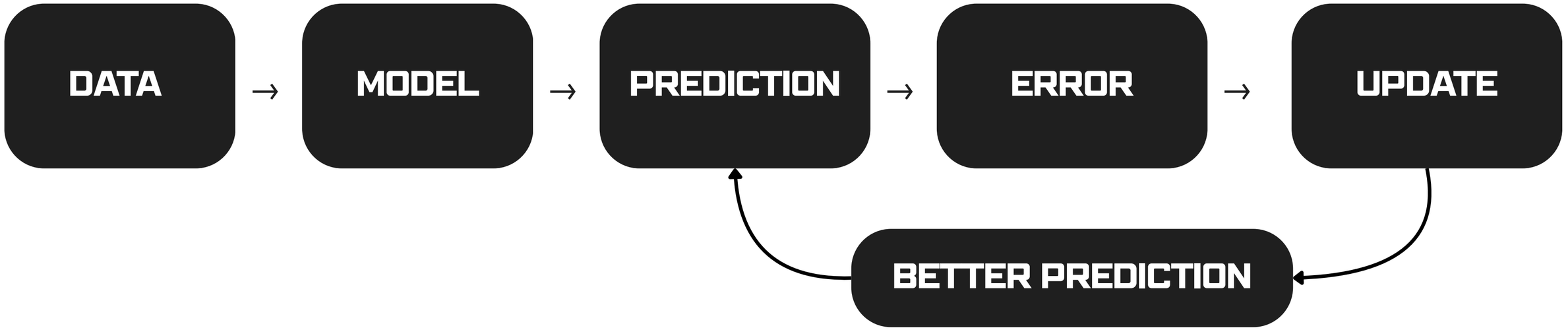
~~4. Linear Algebra for ML~~ ✓

5. Calculus, Probability, and Statistics

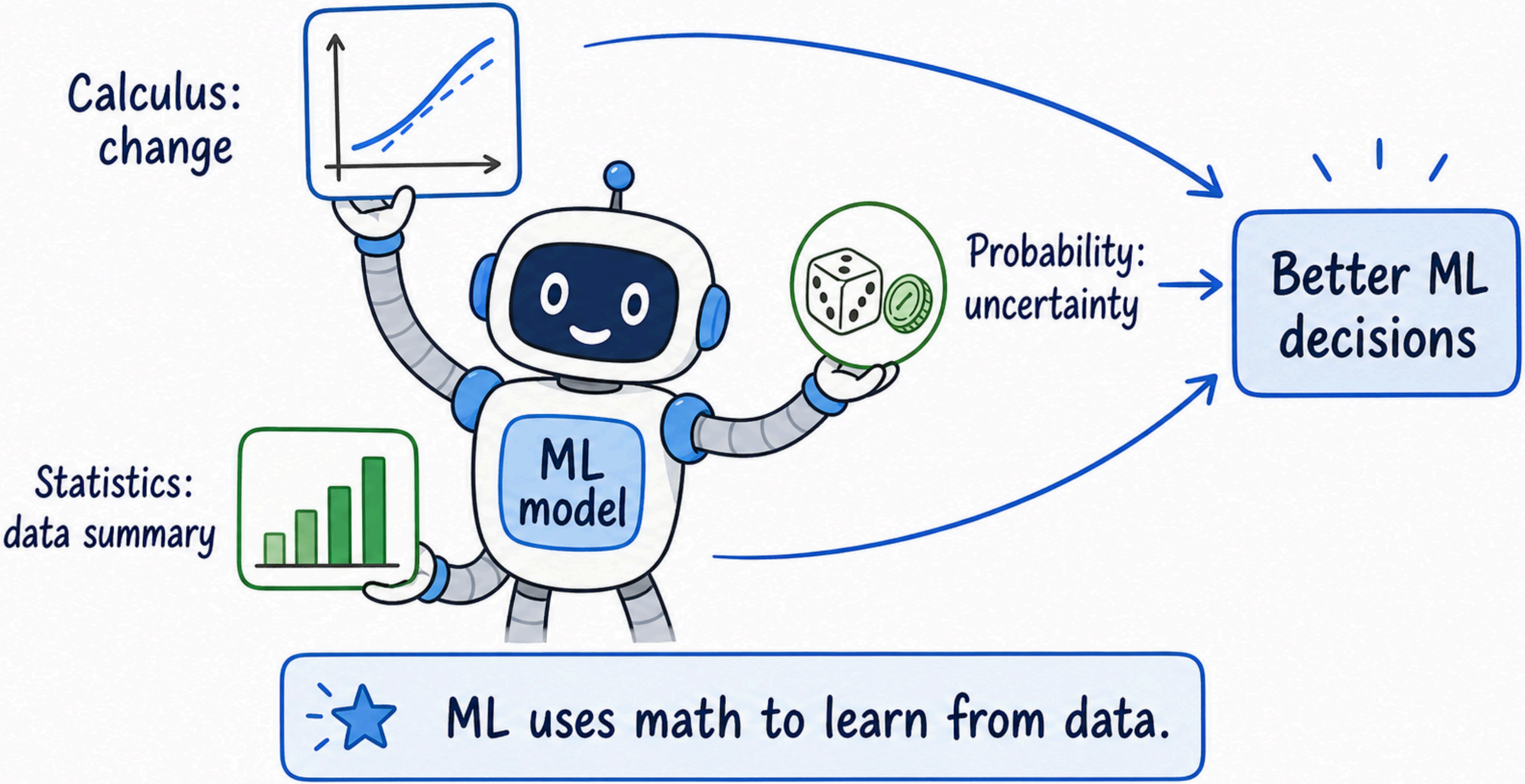
6 LAB

How does a model know how to improve?

ML MODEL'S WORKFLOW



Linear algebra = represent data
Calculus = update model
Probability = express uncertainty
Statistics = understand data


Calculus:
change
Probability:
uncertainty
Statistics:
data summary
ML
model
Better ML
decisions
ML uses math to learn from data.

THE MODEL CAN BE WRONG

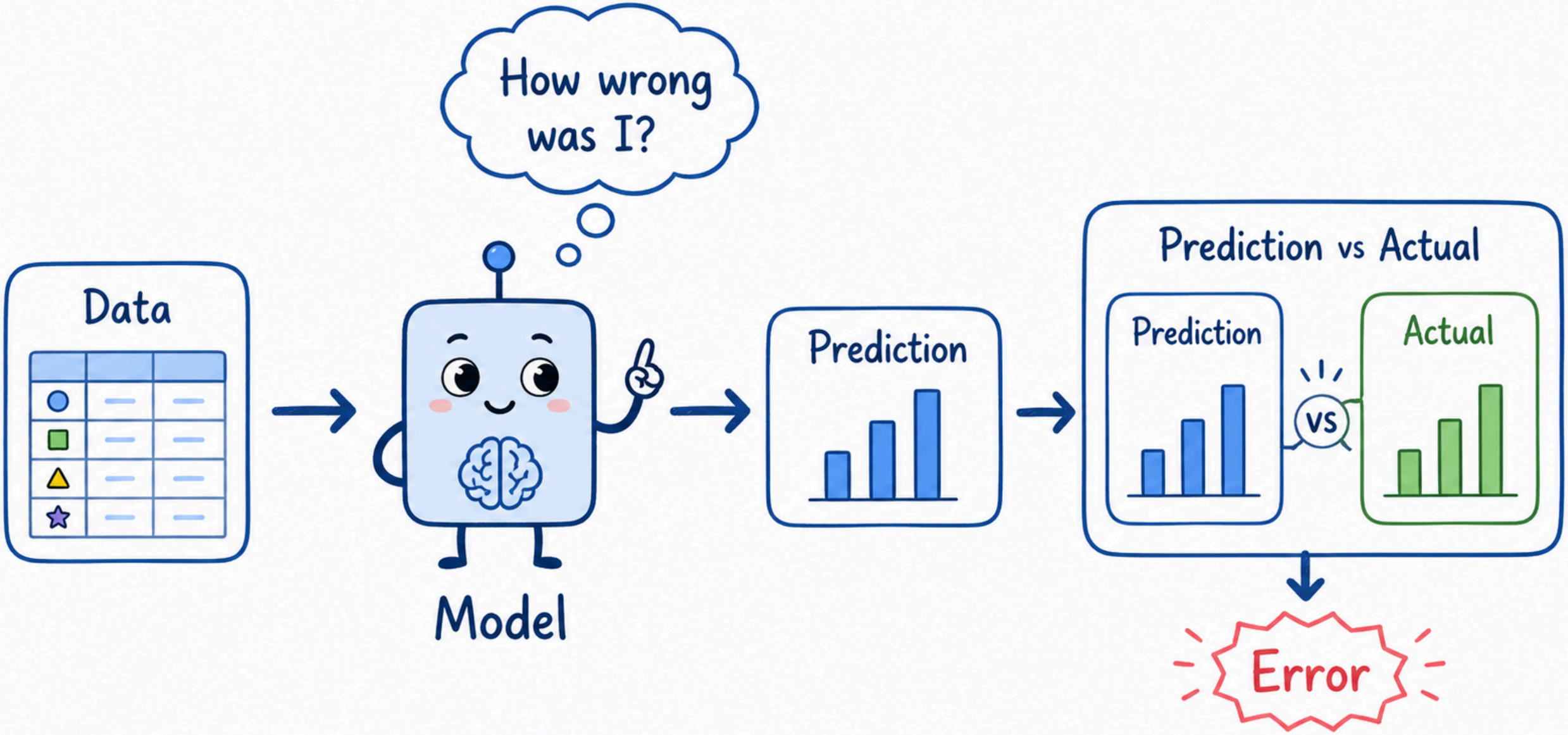
Actual Sales	Predicted Sales	**Error**
100	90	**10**
200	250	**-50**
300	280	**20**

Before improving a model, we need to **measure how wrong it is.**

Error = prediction - actual

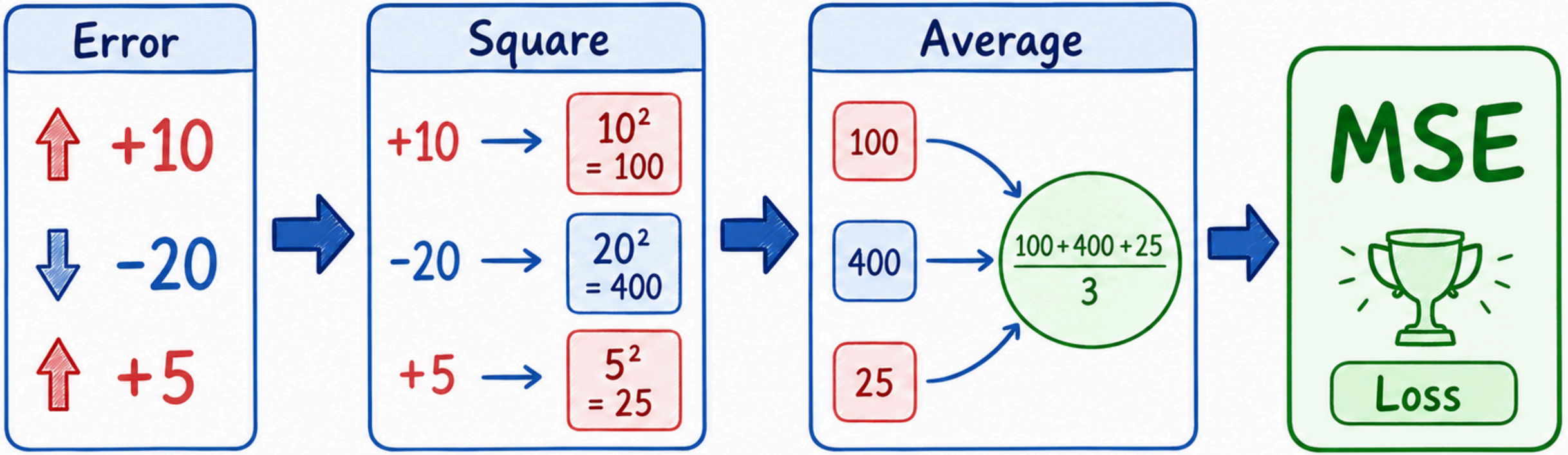
Loss = model errors (of each prediction) summarized into one number

- low loss = better
- high loss = worse



To improve, a model must measure its mistake.

MSE — Mean Squared Error



Positive and negative errors become positive after squaring, then are averaged.

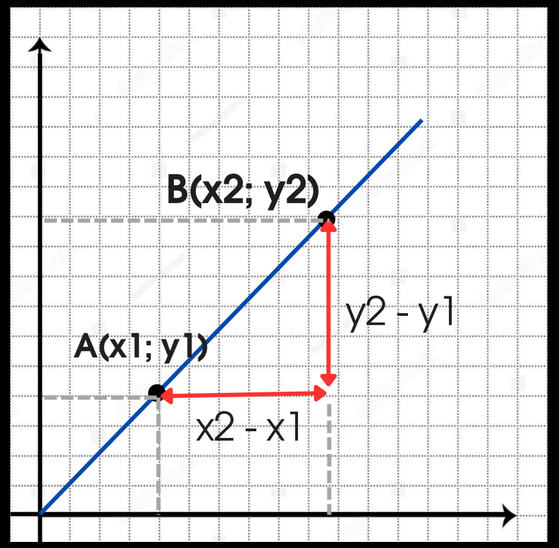
MSE gives one number for how wrong the model is.

SLOPE = CHANGE

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change in parameters → change in predictions | but **how to measure?**

Slope (m) = how much *y* changes per 1 *x* change

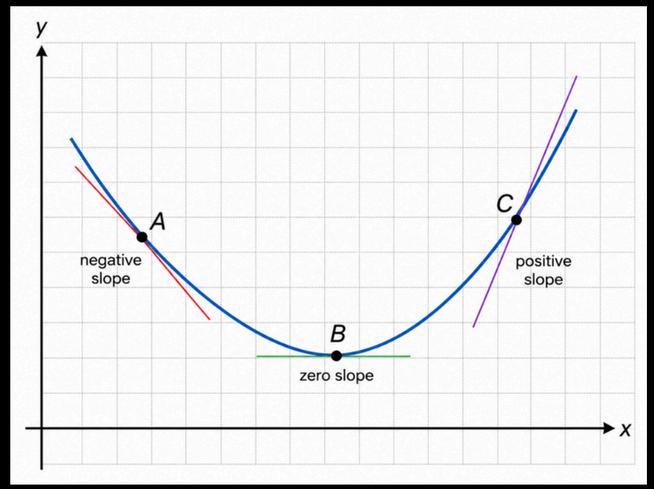


For linear function:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

m > 0 → *x* increases, *y* increases
m < 0 → *x* increases, *y* decreases
m = 0 → *x* changes, *y* doesn't

For curve: **derivative = slope** in a **particular point** of a curve

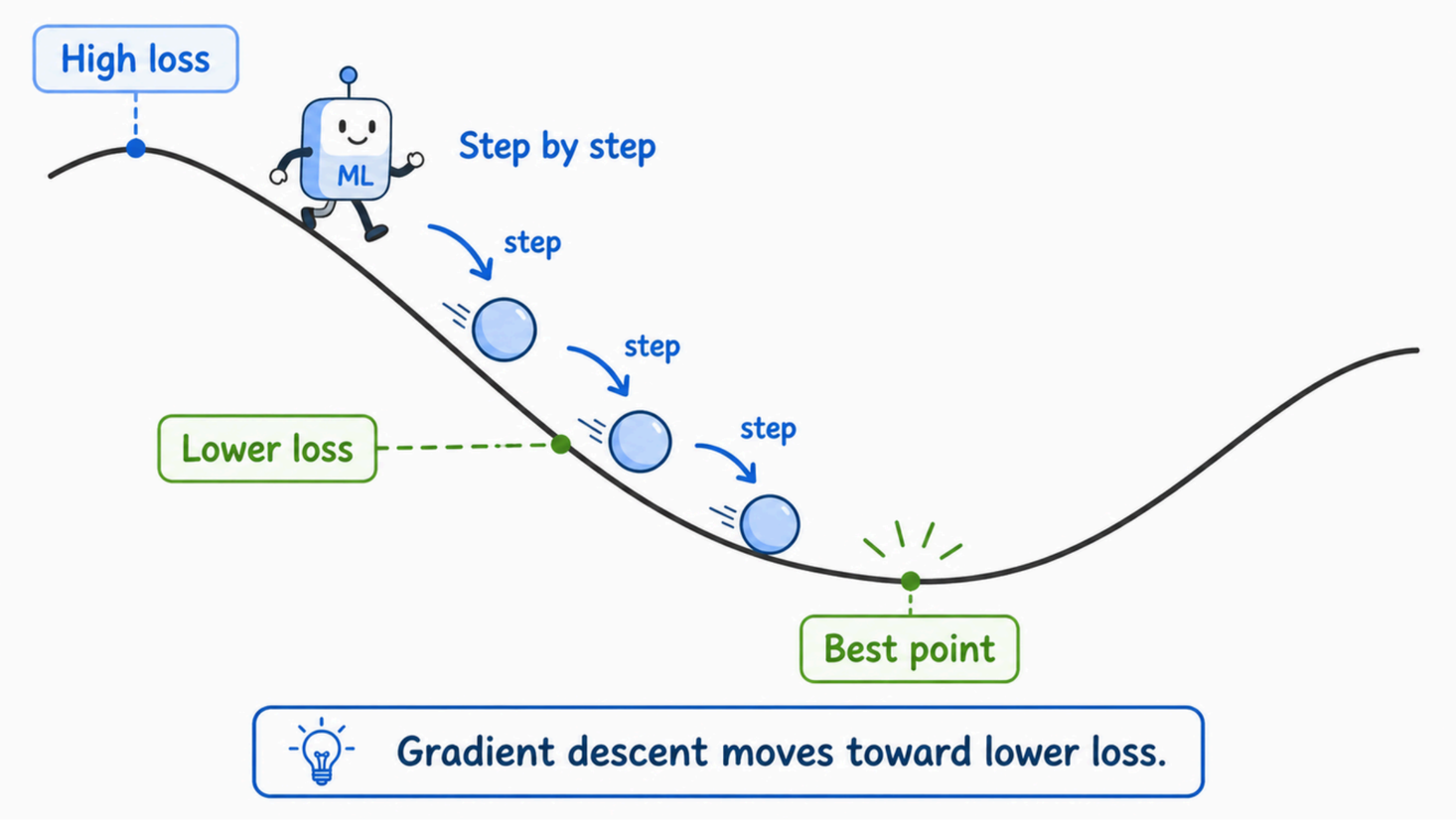


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GRADIENT DESCENT

change in parameters → change in predictions → change in error rates

Goal: find the model parameters that minimize the loss



Gradient descent is an algorithm for **reducing loss**.

Start with **random** parameters, calculate the **slope of the loss**,

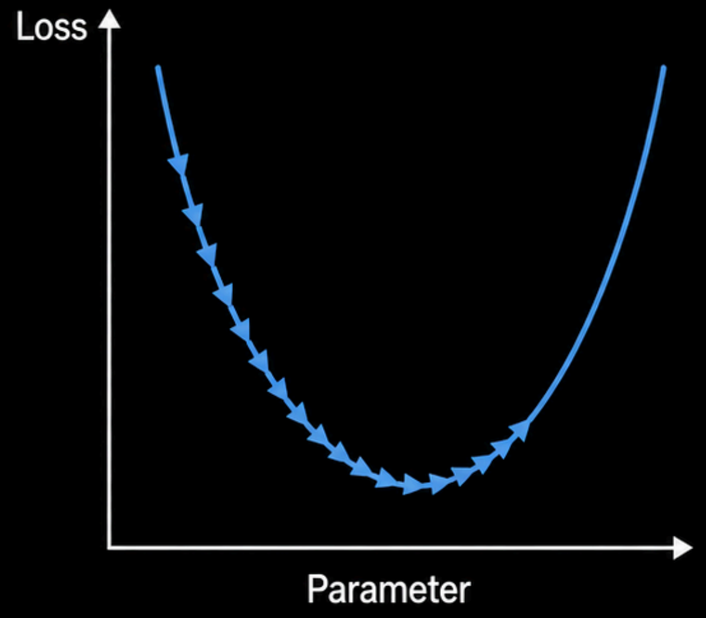
take a small **step downhill,** and **repeat.**

LEARNING RATE

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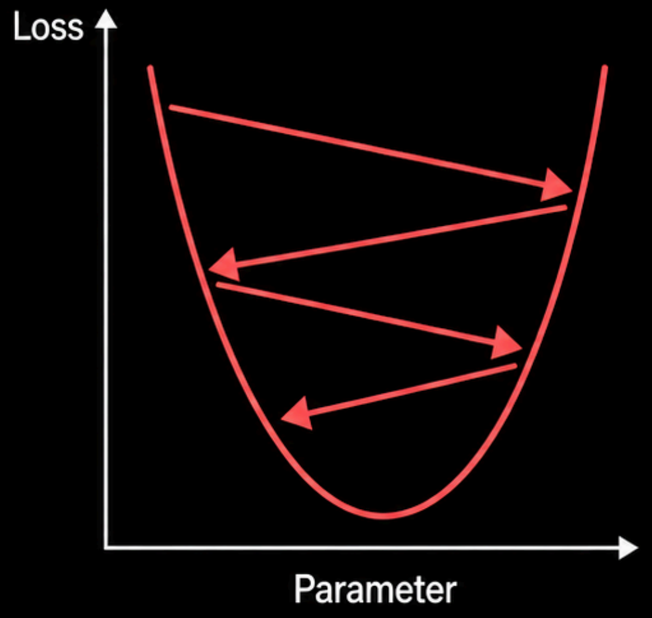
Learning Rate: Its Impact on Optimization

1. Too small learning rate



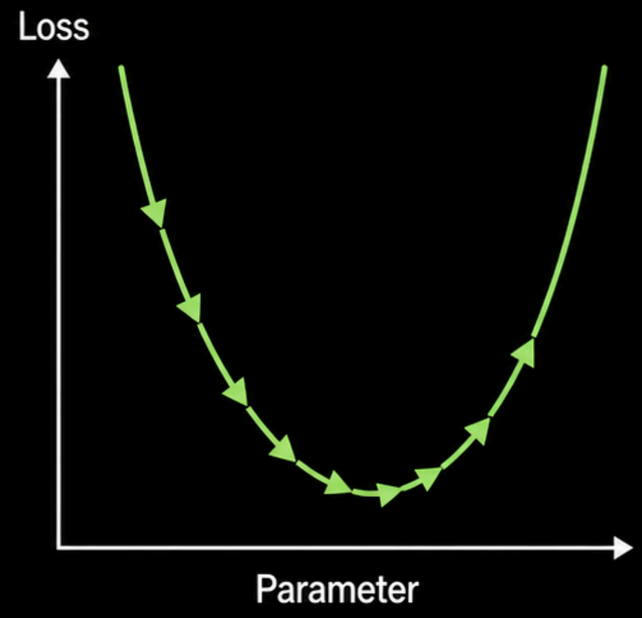
very slow learning

2. Too large learning rate



overshoots the minimum

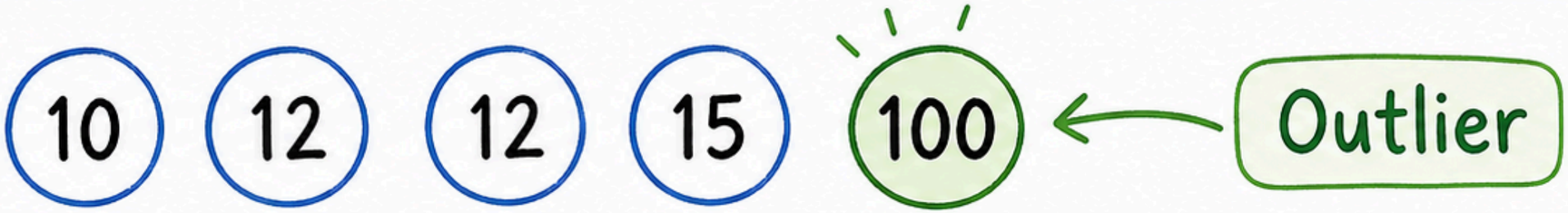
3. Good learning rate



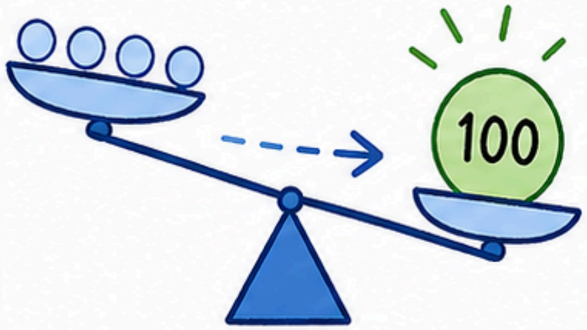
stable movement toward minimum

PROBABILITY

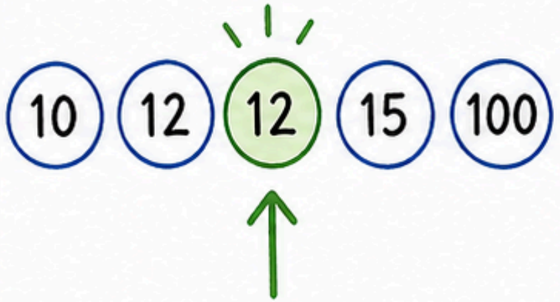
Statistics You Already Know



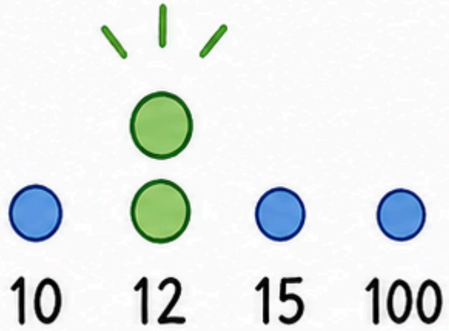
Mean



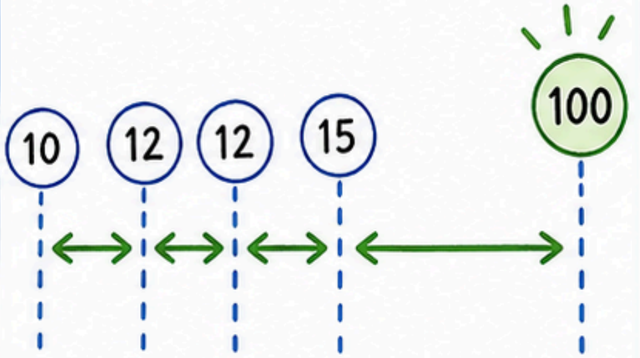
Median



Mode



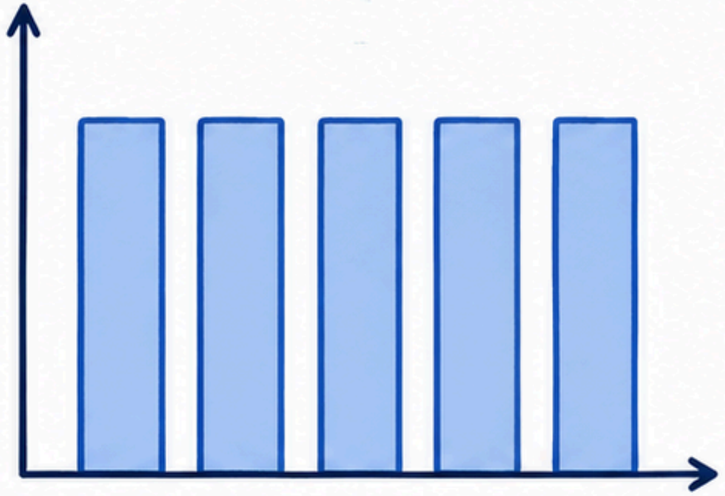
Spread



Statistics helps us summarize messy data.

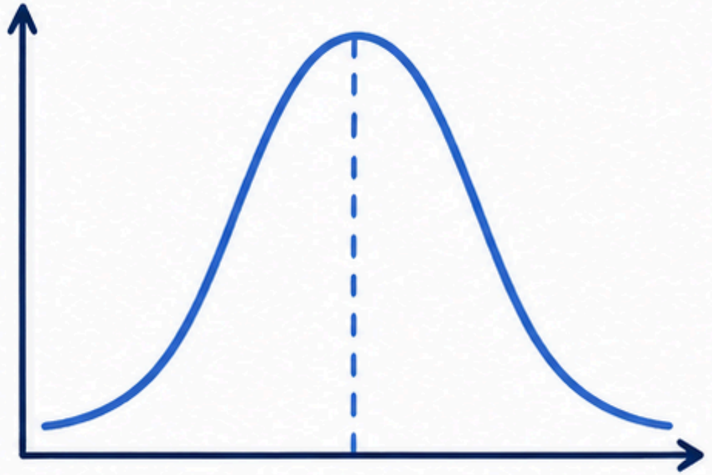
Uniform

Equally likely



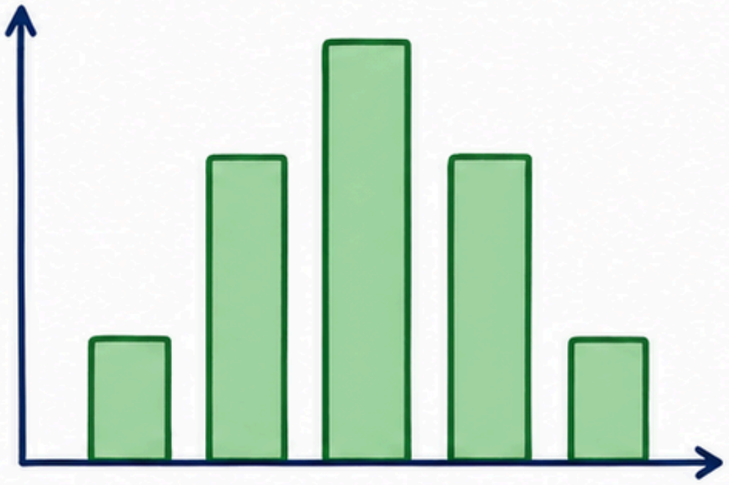
Normal

Around the mean



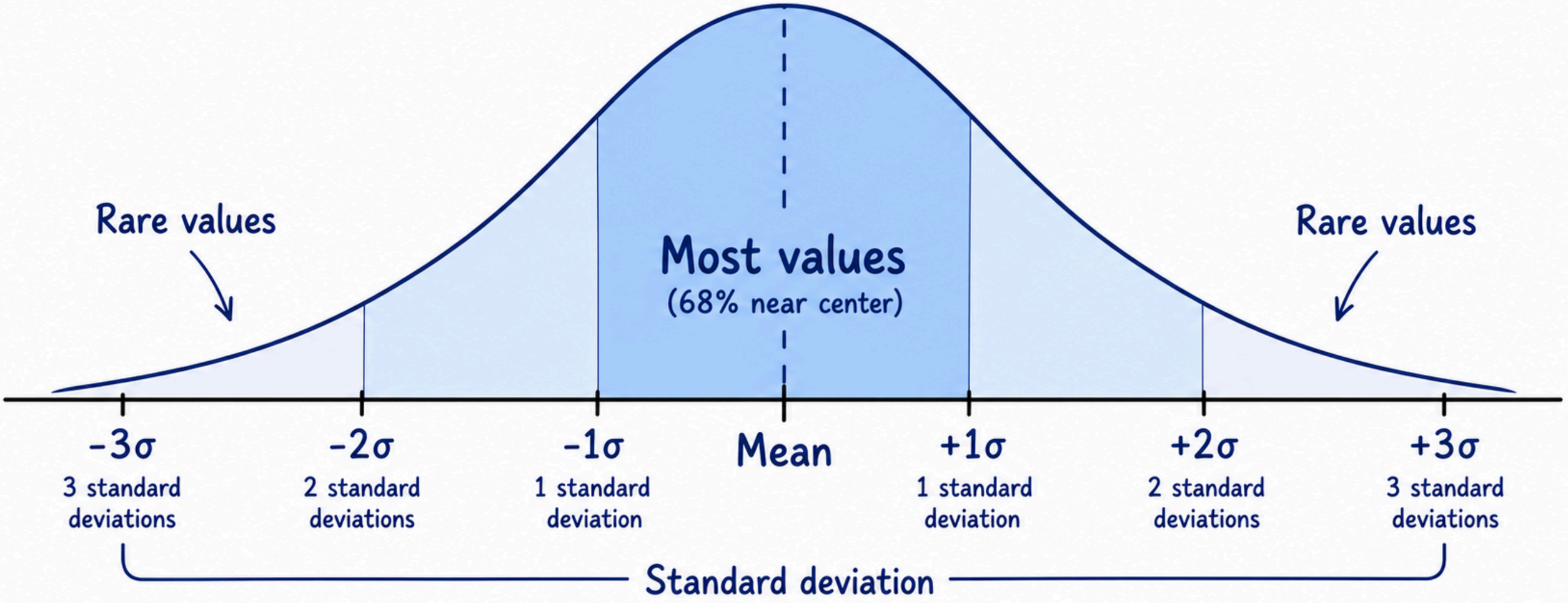
Binomial

Successes



A distribution shows how values are spread.

The Bell Curve



✓ In a normal distribution, most values are near the mean.

CONDITIONAL PROBABILITY

- the probability of one event given that another event already happened.

e.g. customer is more likely to leave, if there were 3 complaints from him

$$P(A|B) = P(B|A) * P(A) / P(B)$$

P(A) = prior belief
P(B | A) = likelihood
P(B) = evidence
P(A | B) = updated belief / posterior

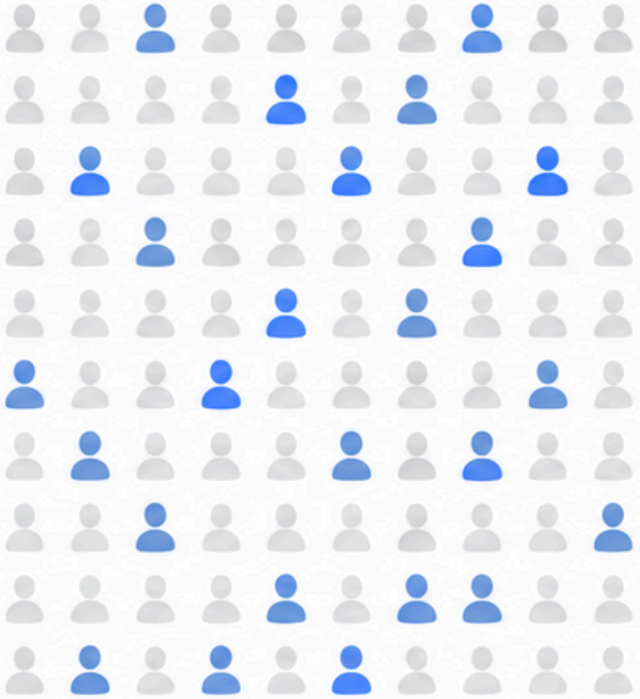
All customers



100

Probability changes with new information

Given: clicked ad



Probability changes with new information

Bought product



Conditional probabillity means probability after we know something.

HYPOTHESIS TESTING

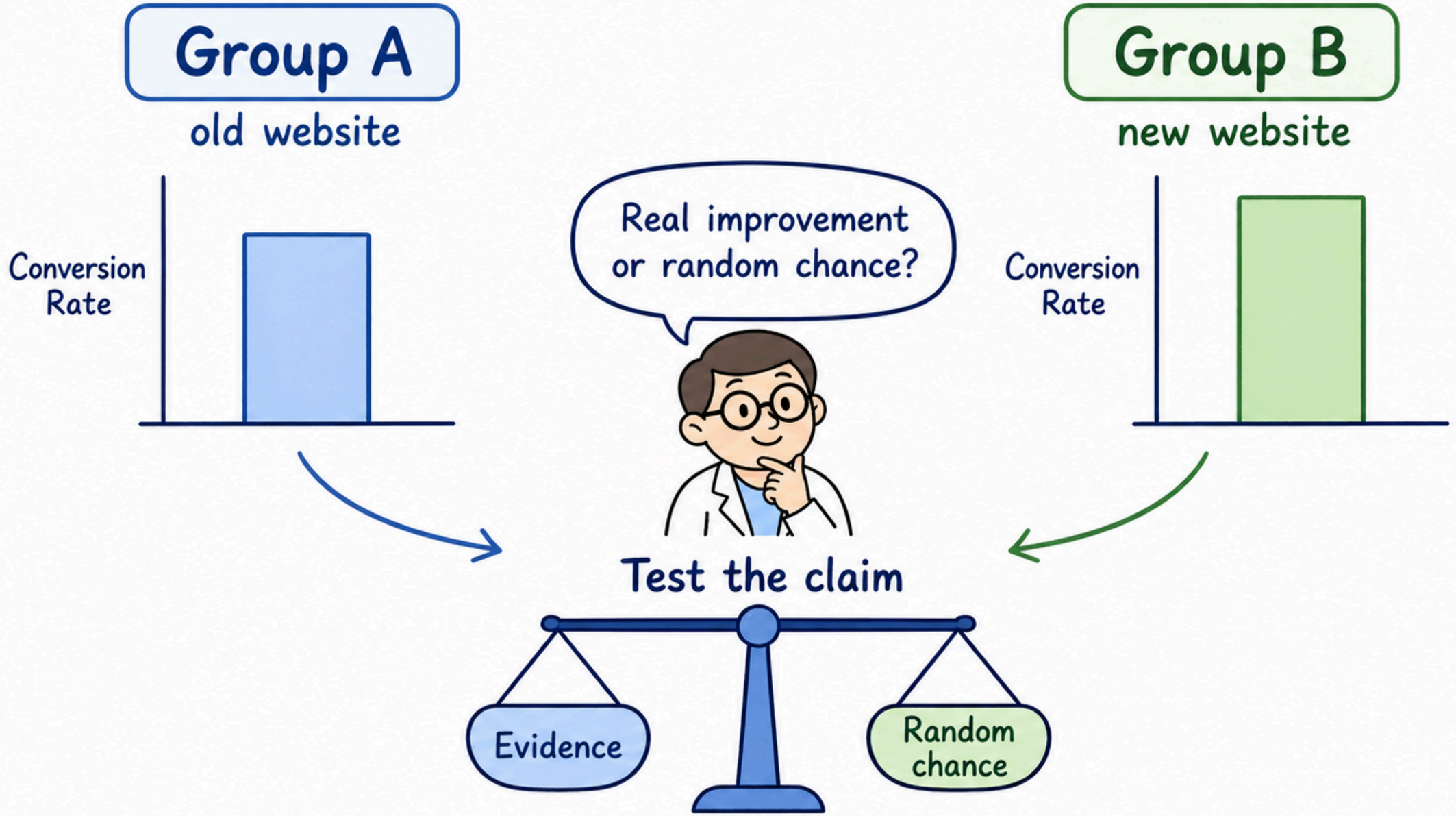
Real effect or random noise?

Model A accuracy = 84%
Model B accuracy = 86%

| Which is better?

H0 / null hypothesis = No real effect until the data gives strong evidence.
H1 / alternative hypothesis = There is the real effect supported by the data

$p < 0.05 \rightarrow$ reject H0 | how likely it would be to see the different result



Hypothesis testing checks if a difference is meaningful.

KEY TAKEAWAYS

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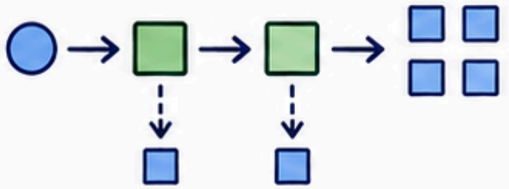
Calculus, Probability, and Statistics

- Distributions — The Shape of Randomness
- The Bell Curve — Why It's Everywhere
- Bayes' Theorem — Updating Beliefs with Evidence
- Next class: stay tuned for Class 6

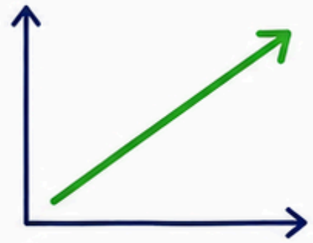
Key Takeaways: Math for ML



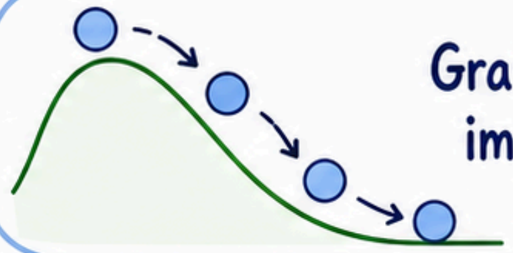
Loss: how wrong



MSE



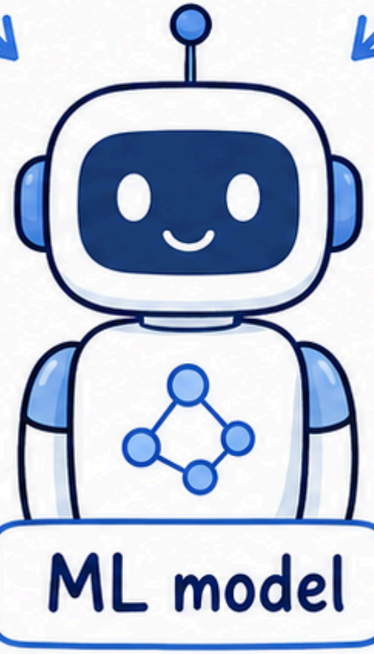
Slope: direction of change



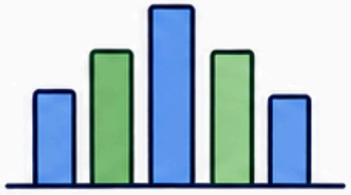
Gradient descent: improve step by step



Probability: uncertainty



ML model



Statistics: summarize data

ML improves by measuring, updating, and reasoning with uncertainty.

THANK YOU

STAY IN TOUCH

HOMEWORK IN THE GROUP!